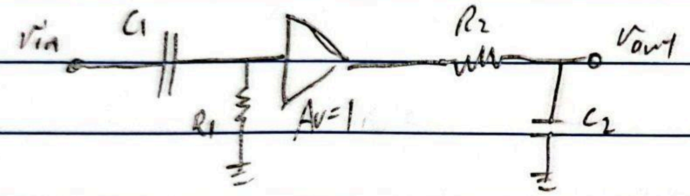
Chai Zi Yang A0262349Y Tutorial 1
Question 1

1)



$R_1 = 10k$ $\quad R_2 = 1k$

$C_1 = 15nF$ $\quad C_2 = 560pF$

i)

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{R_1}{R_1 + \frac{1}{sC_1}} \times 1 \times \frac{\frac{1}{sC_2}}{\frac{1}{sC_2} + R_2}$$

$$= \frac{sR_1C_1}{sR_1C_1 + 1} \times \frac{1}{1 + sR_2C_2}$$

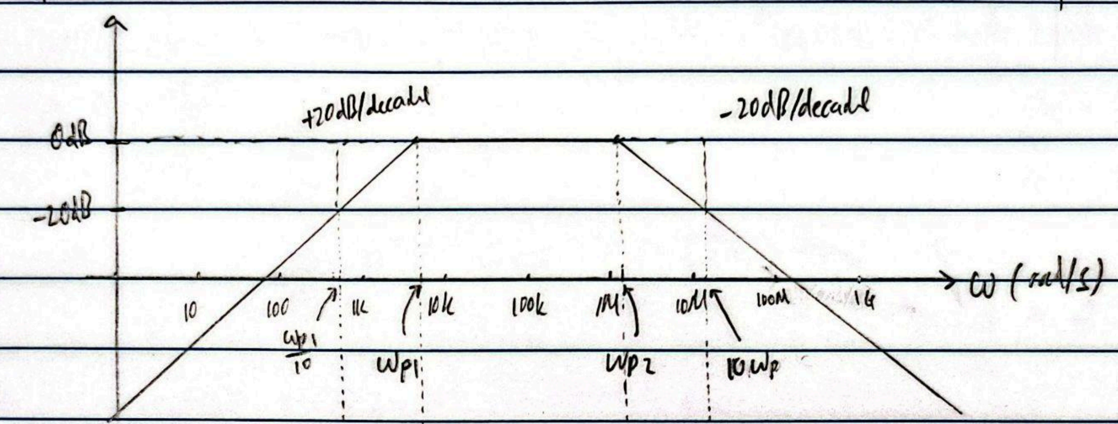
$$= R_1C_1 \frac{s}{\left(\frac{s}{1/R_1C_1} + 1\right)\left(\frac{s}{1/R_2C_2} + 1\right)}$$

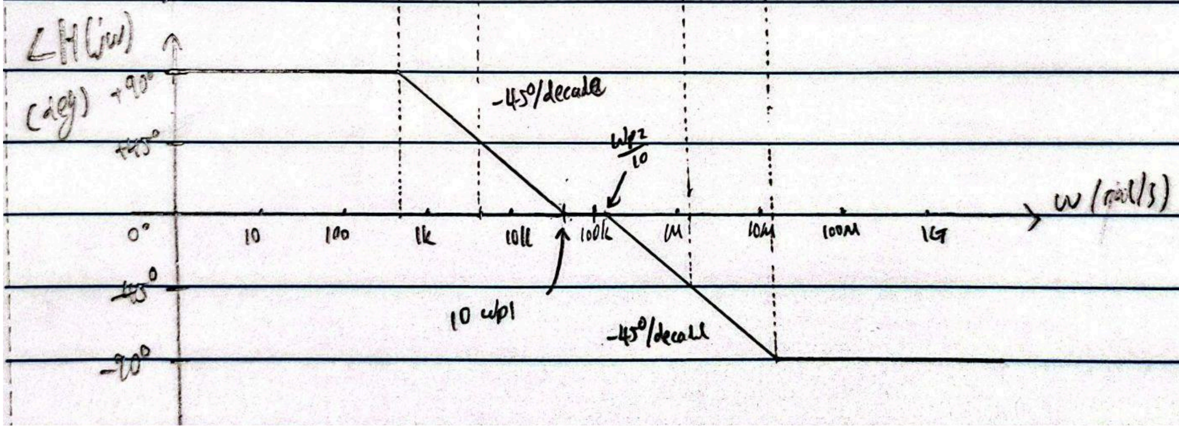
$$= 1.5 \times 10^{-4} \frac{s}{\left(\frac{s}{20000/3} + 1\right)\left(\frac{s}{12500000/7} + 1\right)} \ \#$$

ii) $\omega_z = 0$ rad/s $\quad \omega_{p1} = 20000/3 \approx 6666.67$ rad/s $\quad \omega_{p2} = 12500000/7 \approx 1785714.29$ rad/s

$\Rightarrow f_z = 0Hz \quad f_{p1} \approx 1.061kHz \quad f_{p2} \approx 284.205kHz$ #

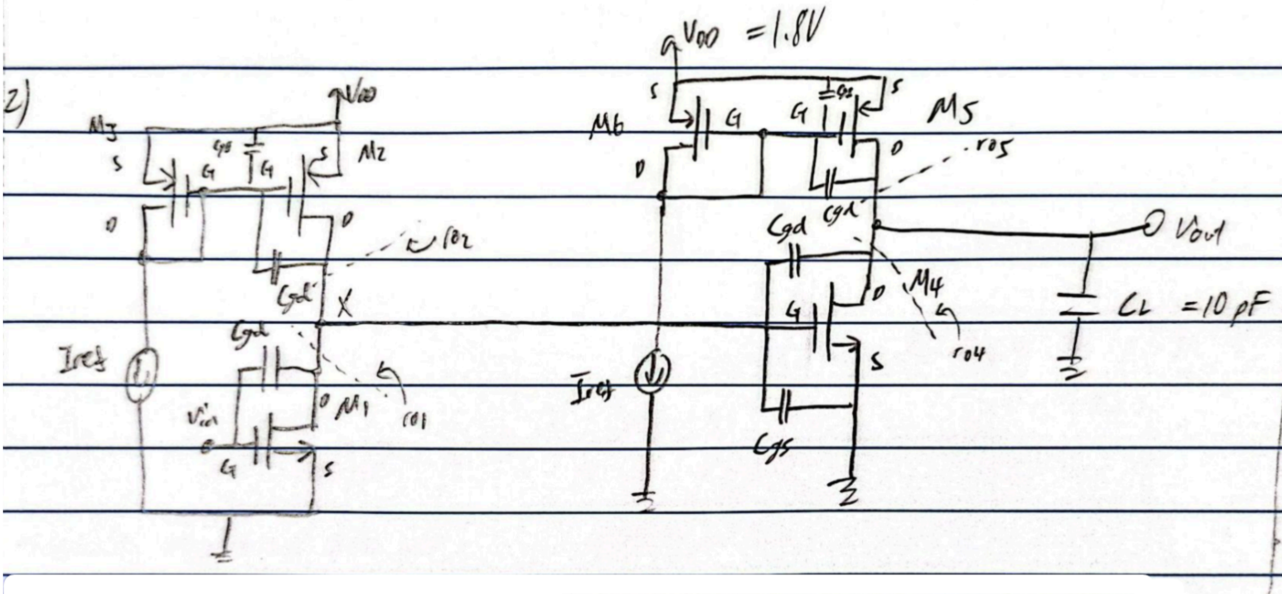
iii) $|H(j\omega)|$ $\qquad \left|H(s)\right|\Big|_{s=j100k} \approx 1$





Question 2

2)



$W/L = \frac{2\mu m}{0.18\mu m}$

$C_{gd} = 10\,fF$

$C_{gs} = 15\,fF$

$V_{TH} = 0.4V$

$\lambda = 0.1\ V^{-1}$

$k_n' = 200\times10^{-6}\ A/V^2$

$k_p' = 100\times10^{-6}\ A/V^2$

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i) $V_{out} = 0.9V,\ \lambda \neq 0 \Rightarrow$ cannot assume all $I_D = I_{ref}$ (there could be gain error)

M_4 & M_5 : $\frac{1}{2}k_p'\left(\frac{W}{L}\right)_5(|V_{GS5}| - V_{TH})^2(1+\lambda_p|V_{DS5}|) = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_4(V_{GS4} - V_{TH})^2(1+\lambda_n V_{DS4})$

$$(|V_{GS5}| - 0.4)^2\cancel{(1+0.1\times0.9)} = 2\,(V_x - 0.4)^2\cancel{(1+0.1\times0.9)}$$

$$(|V_{GS5}| - 0.4)^2 = 2(V_x - 0.4)^2$$

M_6 : $I_{D6} = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_6(|V_{GS6}| - V_{TH})^2(1+\lambda_p|V_{DS6}|)$ $\quad |V_{GS6}| = |V_{DS6}|$

$10\mu = \frac{1}{2}\times100\mu\times\frac{2}{0.18}(\alpha - 0.4)^2(1+0.1\alpha)$ $\quad$ let $|V_{GS6}| = \alpha$

$$0.018 = (\alpha^2 - 0.8\alpha + 0.16)(1+0.1\alpha)$$

$$0.018 = \alpha^2 - 0.8\alpha + 0.16 + 0.1\alpha^3 - 0.08\alpha^2 + 0.016\alpha$$

$$0.1\alpha^3 + 0.92\alpha^2 - 0.784\alpha + 0.142 = 0$$

$\alpha = -9.998\ldots,\quad 0.5307\ldots,\quad 0.2676\ldots$

-ve $\Rightarrow$ NA $\qquad$ $< 0.4V = V_{TH} \Rightarrow$ NA

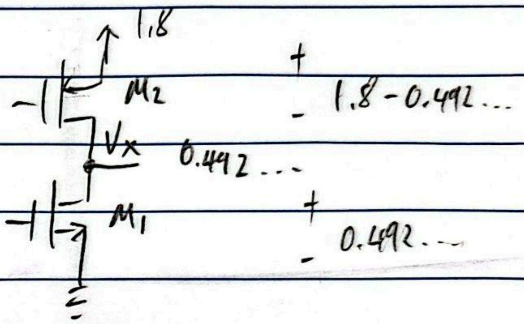
$\therefore |V_{GS6}| = |V_{DS6}| = 0.5307394994\ V$

sub: $2(V_x - 0.4)^2 = (0.5307394994 - 0.4)^2$ $\quad (|V_{GS5}| = |V_{GS6}|)$

$V_x = 0.4924467866\ V$ $\quad$ (ignore negative solution of $\sqrt{\cdot}$)

~~M1 & M2 : from M4 & M5~~

$\cancel{(|V_{GS2}|-0.4)^2 = 2(V_{in}-0.4)^2}$



M_7 : from M_6

$|V_{GS7}| = |_{GS6}| = 0.5307394994\ V$

$\Rightarrow |V_{GS2}| = 0.5307394994\ V$

M_1 & M_2 : $\cancel{\tfrac{1}{2}} k_p' \cancel{\left(\tfrac{W}{L}\right)_2} (|V_{GS2}| - V_{TH})^2 (1+\lambda_p |V_{DS2}|) = \cancel{\tfrac{1}{2}} k_n' \cancel{\left(\tfrac{W}{L}\right)_1} (V_{GS1} - V_{TH})^2 (1+\lambda_n V_{DS1})$

$(|V_{GS2}| - 0.4)^2 (1+0.1|V_{DS2}|) = 2(V_{GS1} - 0.4)^2 (1+0.1 V_{DS1})$

$(0.5307394994 - 0.4)^2 (1+0.1(1.8-V_x)) = 2(V_{in} - 0.4)^2 (1+0.1 V_x)$

$V_{in} = 0.4959704989\ V$ (ignoring -ve solution of $\sqrt{\ }$)

$V_{in} \approx 0.496\ V$ #

ii) current source load common source amplifier (Assume all body terminated is GND for nmos & VDD for pmos)

$\Rightarrow A_V = \frac{v_{out}}{v_{in}} = -g_{m1}(r_{o1} \| r_{o2}) \times -g_{m4}(r_{o4} \| r_{o5})$ $\qquad g_m = 2\sqrt{\tfrac{1}{2} k_{n/p}' \left(\tfrac{W}{L}\right) |I_D|}$

$= g_m^2 \times \left(\frac{r_o}{2}\right)^2$ $\qquad r_o = \frac{1}{\lambda |I_D|}$

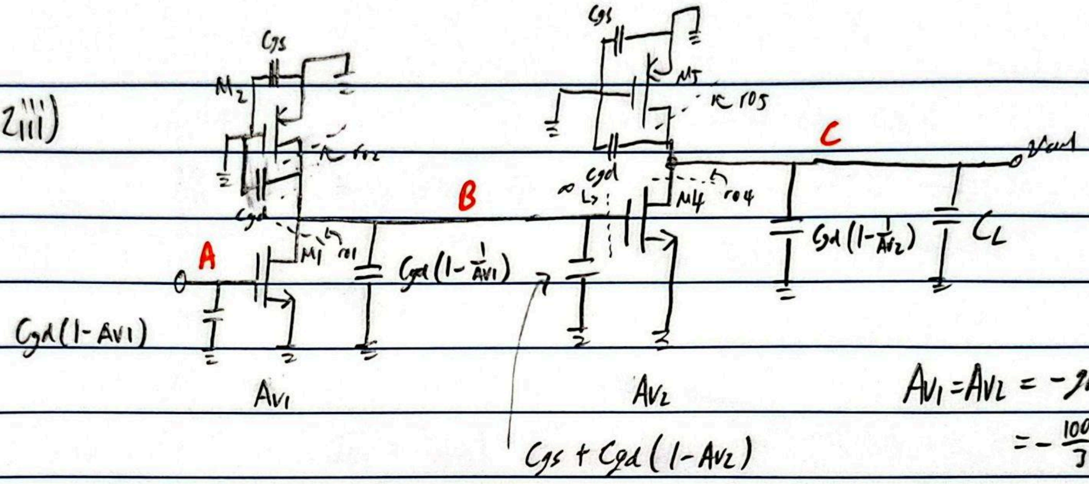
$= 4\left(\frac{1}{2} \times 200\times10^{-6} \times \frac{2}{0.18} \times 10\mu\right) \times \frac{1}{4}\left(\frac{1}{0.1\times 10\mu}\right)^2$

$= 100\,000/9$ $\qquad$ ignoring $\lambda \Rightarrow$ all $I_D = I_{ref}$

$\approx 11111.11\ V/V$ #

$\approx 81\ dB$ #

2iii)



$C_{gd} = 10fF$ $\quad C_L = 10pF$

$C_{gs} = 15fF$

$A_{v1} = A_{v2} = -g_m (r_o/2)$

$= -\frac{100}{3}\sqrt{10}$

Association of poles w/ nodes:

at A: input driven with no R_s => no poles

at B: $R = r_{o1} \| r_{o2}$ $\quad C = C_{gd} + C_{gd}(1 - \frac{1}{A_{v1}}) + C_{gs} + C_{gd}(1 - A_{v2})$

$\approx 500k\Omega$ $\quad \approx 1099.19 fF$

$\omega_{pB} = \frac{1}{RC} \approx 1819521.65$ rad/s # => 289.59 kHz #

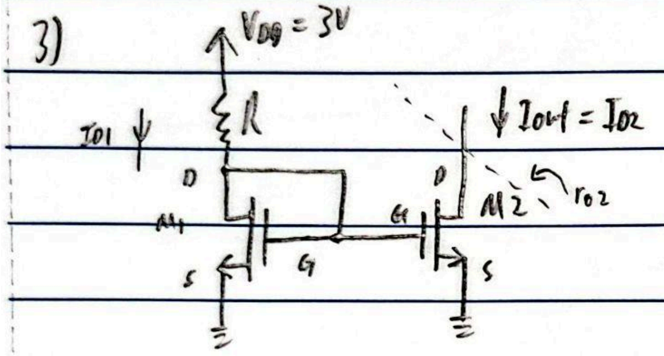
at C: $R = r_{o4} \| r_{o5}$ $\quad C = C_{gd} + C_{gd}(1 - \frac{1}{A_{v2}}) + C_L$

$\approx 500k\Omega$ $\quad \approx C_L = 10pF$

$\omega_{pC} = \frac{1}{RC} \approx 200\,000$ rad/s # => 31.83 kHz #

Question 3

3)



$V_{BS} = 0 \Rightarrow$ no body effect

$L = 1\mu m$ $\quad$ $k_n' = 200 \mu A/V^2$

$V_{TH} = 0.6V$ $\quad$ $\lambda = 0.02\ V^{-1}$

a) $V_{OD} = 0.3V$

$V_{GS1} - V_{TH} = 0.3$

$V_{GS1} = 0.9V$

$V_{DS1} = 0.9V$

$I_{D1} = 15\mu A$

$\frac{3-0.9}{R} = 15\mu A$

$R = 140\ k\Omega$ #

$I_{D1} = \frac{1}{2} k_n' \left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$

$15\mu A = \frac{1}{2} \times 200\mu \times \frac{W_1}{1\mu} \times 0.3^2 \times (1 + 0.02 \times 0.9)$

$W_1 = 2500/1527\ \mu m$

$W_1 \approx 1.64\ \mu m$ #

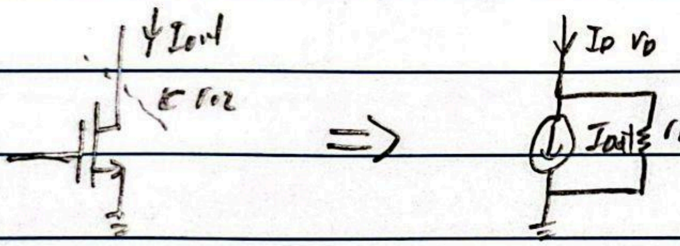
b) Assume gain error free $V_{D2} = V_{D1}$

$$I_{out} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} I_{REF}$$

$$= \frac{W_2}{W_1} \times 15\mu A$$

I_{out}, μA	7.5	30	45	
W_2, μm	$0.5W_1$	$2W_1$	$3W_1$	
	≈ 0.82	≈ 3.28	≈ 4.92	#
R_{out}, $M\Omega$	6.67	1.67	1.11	#

c) R_{out} of current source $= r_o = \frac{1}{\lambda I_D}$



How I_o changes w/ V_o $\Rightarrow$ $\frac{\partial I_o}{\partial V_o} = \frac{\partial}{\partial V_o}\left(I_{out} + \frac{V_o}{r_{o2}}\right)$

$= \frac{1}{r_{o2}} = \frac{1}{1.67M\Omega} \approx 0.6\ \mu A/V$

if V_o is doubled from for example 1V to 2V,

I_{out} increased from $30\mu A + 0.6\mu A$ to $30\mu A + 1.2\mu A$

$\Rightarrow$ I_{out} changes $\approx 0.6\mu A$ for every 1V in V_o #